Chapter 5: Python Operators (`operators.py`)

Type	Draft
Status	Not started
Project	My Python Journey

1. What are operators?

Operators are special symbols or keywords used to perform operations on variables and values.

Example:

```
a = 10b = 5print(a + b)
```

Output:

```
15
```

Here, `+` is an operator.

2. Types of Operators in Python

Python has 7 main categories of operators.

No.	Operator	Purpose
1	Arithmetic	Mathematical calculations
2	Comparison	Compare two values
3	Assignment	Assign values to variables
4	Logical	Combine conditions

No.	Operator	Purpose
5	Bitwise	Perform operations on binary numbers
6	Membership	Check whether a value exists
7	Identity	Check whether two references point to the same object

3. Arithmetic Operators

Used to perform mathematical calculations.

Operator	Name	Example	Result
`+`	Addition	`10 + 5`	15
`-`	Subtraction	`10 - 5`	5
`*`	Multiplication	`10 * 5`	50
`/`	Division	`10 / 5`	2.0
`//`	Floor division	`10 // 3`	3
`%`	Modulus	`10 % 3`	1
`**`	Exponentiation	`2 ** 3`	8

Example:

```
a = 10b = 3print(a + b)print(a - b)print(a * b)print(a / b)print(a // b)print(a % b)print
(a ** b)
```

Output:

```
13
7
30
3.3333333333333335
3
1
1000
```

Important:

- `/` gives division result as a float.

- `//` gives floor division.
- `%` gives remainder.
- `*` calculates power.

4. Comparison Operators

Used to compare two values. The result is always Boolean (`True` or `False`).

Operator	Meaning	Example
`==`	Equal to	`10 == 10` → True
`!=`	Not equal to	`10 != 5` → True
`>`	Greater than	`10 > 5` → True
`<`	Less than	`5 < 10` → True
`>=`	Greater than or equal to	`10 >= 10` → True
`<=`	Less than or equal to	`5 <= 10` → True

Example:

```
a = 10b = 5print(a == b)print(a != b)print(a > b)print(a < b)
```

Output:

```
False
True
True
False
```

5. Assignment Operators

Used to assign values to variables and update their values.

Operator	Example	Equivalent
`=`	`a = 10`	Assign 10
`+=`	`a += 5`	`a = a + 5`

Operator	Example	Equivalent
`-=`	`a -= 5`	`a = a - 5`
`*=`	`a *= 5`	`a = a * 5`
`/=`	`a /= 5`	`a = a / 5`
`//=`	`a //= 5`	`a = a // 5`
`%=`	`a %= 5`	`a = a % 5`
`**=`	`a **= 2`	`a = a ** 2`

Example:

```
a = 10a += 5print(a)a *= 2print(a)
```

Output:

```
15
30
```

6. Logical Operators

Used to combine multiple conditions.

Operator	Meaning
`and`	True when both conditions are true
`or`	True when at least one condition is true
`not`	Reverses the Boolean result

Example:

```
a = 10print(a > 5 and a < 20)print(a > 15 or a < 20)print(not(a > 5))
```

Output:

```
True
True
```

```
False
```

7. Bitwise Operators

Bitwise operators perform operations on the binary representation of numbers.

Operator	Name
`&`	AND
`|`	OR
`^`	XOR
`~`	NOT
`<<`	Left shift
`>>`	Right shift

Example:

```
a = 5b = 3print(a & b)print(a | b)print(a ^ b)print(a << 1)print(a >> 1)
```

Output:

```
1
7
6
10
2
```

8. Membership Operators

Used to check whether a value exists in a sequence or collection.

Two membership operators:

- `in`
- `not in`

Example:

```
fruits = ["apple", "mango", "banana"]print("apple" in fruits)print("grapes" not in fruit
s)
```

Output:

```
True
True
```

9. Identity Operators

Used to check whether two variables refer to the same object in memory.

Two identity operators:

- `is`
- `is not`

Example:

```
a = [10, 20]b = ac = [10, 20]print(a is b)print(a is c)print(a == c)
```

Output:

```
True
False
True
```

Remember the difference:

- `==` compares values.
- `is` compares object identity.

10. Important differences for interviews

Operators	Difference
`/` and `//`	Normal division vs floor division

Operators	Difference
`=` and `==`	Assignment vs comparison
`==` and `is`	Value equality vs object identity
`and` and `or`	Both conditions vs at least one condition
`%` and `//`	Remainder vs floor quotient

11. Important practice questions

Try these programs before checking answers.

Basic coding questions

1. Write a program to perform addition, subtraction, multiplication and division.
2. Write a program to find the square and cube of a number.
3. Write a program to check whether a number is even or odd using `%`.
4. Write a program to calculate the remainder and quotient of two numbers.
5. Write a program to compare two numbers.
6. Write a program to check voting eligibility using logical operators.
7. Write a program to check whether a particular fruit exists in a list.
8. Write a program to swap two numbers using assignment.

Tricky output questions

Q1.

```
print(10 // 3)print(10 / 3)
```

Q2.

```
x = 5x += 3x *= 2print(x)
```

Q3.

```
print(5 > 3 and 10 < 5)
```

Q4.

```
print(2 ** 3 ** 2)
```

Q5.

```
a = [1, 2]b = [1, 2]print(a == b)print(a is b)
```

12. GitHub upload

Your file structure can look like this:

```
python-learning/
│
├── python_introduction.py
├── variables.py
├── data_types.py
├── type_conversion.py
└── operators.py
```

Save the theory as comments and the examples as executable Python code.

Use these commands:

```
git add operators.py
git commit -m "Add Python operators notes and examples"
git push origin main
```

Your next chapter: Conditional Statements (`if`, `elif`, `else`).